

# Numeric Attribute Types

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- Quantity (integer or real-valued)

- Interval

- Measured on a scale of **equal-sized units**
- Values have order
  - E.g., *temperature in C° or F°, calendar dates*
- No true zero-point

- Ratio

- Inherent **zero-point**
- We can speak of values as being an order of magnitude larger than the unit of measurement (10 K° is twice as high as 5 K°).
  - e.g., *temperature in Kelvin, length, counts, monetary quantities*

# Discrete vs. Continuous Attributes

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## ❑ Discrete Attribute

- ❑ Has only a finite or countably infinite set of values
  - ❑ E.g., zip codes, profession, or the set of words in a collection of documents
- ❑ Sometimes, represented as integer variables
- ❑ Note: Binary attributes are a special case of discrete attributes

## ❑ Continuous Attribute

- ❑ Has real numbers as attribute values
  - ❑ E.g., temperature, height, or weight
- ❑ Practically, real values can only be measured and represented using a finite number of digits
- ❑ Continuous attributes are typically represented as floating-point variables